

Coding & Solution

Date	04 JULY 2026
Team ID	SWTID-2026-2662
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/nakkahemasaisree-cmd/AI-powered-debt-relief-and-financial-recovery-platform
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	FastAPI, Streamlit, SQLAlchemy, Pydantic, Uvicorn, Google Gemini API (google-generativeai), Requests
Key Features Implemented	• User Registration & Secure Login using JWT Authentication • Financial Profile Management (Income, Expenses, Lump Sum) • Loan Management (Add, View, Delete Loans) • Financial Health Analysis (EMI Ratio, Debt-to-Income Ratio, Stress Level) • AI-Based Loan Settlement Prediction • AI-Powered Debt Negotiation Strategy Generation using Google Gemini API • Automatic Negotiation Email/Letter Generation • Debt Repayment Timeline Simulator • Dashboard with Financial Statistics • AI History Tracking for Generated Strategies and Letters • Responsive Stream-lit User Interface with Modern Dashboard
Pending / Incomplete Features	• Email Sending Integration (SMTP) • Cloud Deployment (AWS/Azure/Render) • Real-time Bank API Integration • Machine Learning Model Training using Historical Loan Data • Credit Score Prediction Module • Multi-language Support • Admin Dashboard • Notification & Reminder System • Mobile Application Support
Setup / Run Instructions	1. Install Python 3.10+. 2. Install dependencies using <code>pip install -r requirements.txt</code>. 3. Configure the <code>.env</code> file with <code>GOOGLE_API_KEY</code>, <code>DATABASE_URL</code>, <code>SECRET_KEY</code>, and <code>API_BASE_URL</code>. 4. Start the Fast-API back-end using <code>uvicorn app.main:app --reload</code>. 5. Launch the Stream lit frontbench using <code>stream lit run stream lit_app/app.py</code>. 6. Open the application in a browser, register/login, add loan details, and use the dashboard, financial health analysis, settlement predictor, and AI negotiation features.

Code Quality Checklist

S. No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes*

Additional Notes / Comments:

The project follows a well-structured modular architecture using Fast-API for the back-end and Stream-lit for the front end. The source code is organized into separate modules for models, routers, services, scheme, and UI pages, improving maintainability and scalability. JCT-based authentication, exception handling, Alchemy ORM, and Google Gemini AI integration are implemented effectively. The application provides features such as financial health analysis, loan management, AI-powered settlement prediction, negotiation strategy generation, debt repayment simulation, and automated negotiation letter generation. Overall, the project demonstrates good coding practices, modular design, and is suitable for academic submission.